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Prof. Rodney Van Meter, Editor-in-Chief
IEEE Transactions on Quantum Engineering (IEEE TQE)
IEEE Quantum Initiative • IEEE Computer Society • IEEE Photonics Society

SUBJECT: Submission of Original Research Manuscript — "Orbit-Tied Quantized Neural Belief Propagation for Real-Time Sliding-Window Decoding of Bivariate-Bicycle Quantum Codes"

Dear Prof. Van Meter and the Editorial Board of *IEEE Transactions on Quantum Engineering*,

On behalf of my co-authors, I am pleased to submit our original research manuscript, titled **"Orbit-Tied Quantized Neural Belief Propagation for Real-Time Sliding-Window Decoding of Bivariate-Bicycle Quantum Codes,"** for peer-reviewed publication as a Regular Paper in the *IEEE Transactions on Quantum Engineering*. The manuscript consists of 12 pages in standard IEEE Transactions two-column format, comprising 7 multi-panel publication figures, 5 comparative benchmark tables, formal mathematical proofs, synthesizeable hardware designs, and 59 comprehensive citations.

Authorship Sequence & Roles:
• **Md. Nazmul** (Lead & Corresponding Author; ORCID: 0009-0001-6115-7023; E-mail: nazmulhas36@gmail.com)
Role: Lead Researcher, Conceptualization, Methodology & Software Architecture
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Background and Motivation: Quantum Low-Density Parity-Check (QLDPC) bivariate-bicycle (BB) codes, pioneered by Bravyi et al. (Nature 2024), represent a transformative milestone toward scalable fault-tolerant quantum computation, enabling an order-of-magnitude reduction in physical qubit overhead compared to standard 2D topological surface codes. However, translating BB codes into operational quantum processors faces a critical decoding bottleneck: their dense, non-local Tanner graphs produce complex 3D space-time Detector Error Models (DEM) with thousands of correlated fault mechanisms. Conventional decoders (such as BP+OSD) require hundreds of microseconds on high-end CPUs, causing catastrophic backlog and buffer overflow under the strict ~1 μs coherence budget of superconducting quantum processors.

Key Scientific & Engineering Contributions:

1. Torus Automorphism Equivariance (Theorem 2): We prove that the 2D torus translation group $G = Z_\ell \times Z_m$ forms an exact automorphism group of the space-time DEM hypergraph. By tying neural BP weights across automorphism orbits, we collapse 7,308 DEM fault mechanisms into 203 orbits (36.0× reduction) on [[72,12,6]] and 14,616 mechanisms into 203 orbits (72.0× reduction) on [[144,12,12]], guaranteeing mathematical equivariance and eliminating untied neural overfitting.

2. Multiplier-Free 6-Bit Dyadic CSD Architecture (Theorem 3): To eliminate costly Digital Signal Processing (DSP) multiplier blocks entirely (DSP=0), we prove dyadic fixed-point bounds ([-7.75, +7.75], Δ=0.25) and implement all variable- and check-node updates using two-term Canonical Signed-Digit (CSD) shift-and-add arithmetic, enabling ultra-compact FPGA deployment.

3. Multi-Stage Margin Gating & Streaming Sliding Window: We engineer a confidence-margin fast-path gate (G_3) within a continuous sliding window (W=3, F=1). Gate G_3 routes 75.6% of syndrome windows with zero logical errors ($P_{L,accept} = 0.0000$), reducing queue backlog to 6.0×10^{-5} and delivering an end-to-end pipeline logical error rate of $P_{pipe} = 1.90 \times 10^{-4}$ with 102.4 ns mean latency (240 ns max) — well within the 1 µs quantum cycle limit.

4. 100-Round Continuous Streaming Memory Survival: We validate continuous operation across R=100 measurement rounds (100 µs at 1 µs/cycle). Active temporal boundary residual propagation preserves 98.1% logical survival at 100 rounds ($F_{log} = 0.9981$ at 10 µs, 0.9812 at 100 µs), whereas boundary-agnostic decoding collapses to 22.1% (4.4× degradation).

5. Empirical IBM Quantum Heron r2 Calibration Benchmark: Evaluating our decoder against the empirical calibration snapshot of IBM's flagship 133-qubit Heron r2 processor ($p_{2Q} = 2.1 \times 10^{-3}$, $\eta = 50.0$, $p_{RO} = 8.0 \times 10^{-3}$) confirms convergence in only 3.85 mean iterations — the fastest among all 6 models tested (vs. 9.16 for Classical Min-Sum, 11.86 for Normalized Min-Sum, and 8.86 for BP+OSD-CS-10) with DSP=0.

6. Hardware Synthesis & 10,000-Shot Co-Simulation Parity: We synthesize our complete architecture in SystemVerilog for AMD UltraScale+ XCKU040 FPGA ($f_{clk} = 250$ MHz, WNS = +0.18 ns, 28,204 LUTs, 0 DSPs), confirming 10,000-shot bit-exact hardware-software co-simulation parity.

Scope Alignment with IEEE Transactions on Quantum Engineering:

This work addresses the intersection of quantum error-correcting codes, neural message-passing decoders, and real-time FPGA microarchitectures. It directly advances IEEE TQE's core mandate of bridging theoretical quantum information science and scalable physical quantum hardware engineering.

Editorial & Ethical Declarations:

1. **Originality:** This manuscript represents original research and is not currently published, accepted, or under concurrent evaluation in any other journal or conference.
2. **Software & Data Availability:** In accordance with IEEE open science standards, the complete implementation is publicly available as an open-source Python library (`orbit-tied-qnbp` v1.0.0 on PyPI and GitHub: `github.com/NazmulHasanNihal/orbit-tied-qnbp-sliding-window-bb`), including all synthesizeable SystemVerilog RTL, testbenches, and dataset generation scripts.
3. **Conflict of Interest:** The authors certify that there are no financial or commercial conflicts of interest.

Suggested Expert Reviewers:

Reviewer	Institution & Contact	Expertise & Rationale
Dr. Sergey Bravyi	IBM Quantum, T.J. Watson Research `sbravyi@us.ibm.com`	Pioneer of Bivariate-Bicycle QLDPC codes and gross codes.
Dr. Nicolas Delfosse	IonQ (formerly Microsoft Quantum) `ndelfosse@ionq.com`	Expert in sliding-window decoding, Union-Find, and LDPC Tanner graph symmetries.
Dr. Joschka Roffe	Freeform Labs / Univ. of Strathclyde `joschka@roffe.eu`	Developer of BP-OSD and expert in belief propagation on quantum codes.
Prof. Barbara Terhal	Delft University of Technology / QuTech `b.m.terhal@tudelft.nl`	Leader in fault-tolerant architectures and real-time QEC decoding limits.

Thank you for your time, consideration, and management of the peer-review process.

Sincerely,

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